

2023 Ph H2 Q8

Section: Particles and Waves

Topic: The Inverse Square Law (Irradiance)

Summary of Question:

Design and describe an experiment to verify the inverse square law for light. Given irradiance–distance data, (i) define irradiance, (ii) use all data to establish the I–d relationship, and (c) explain physically why irradiance decreases with distance.

(a) Describe an experiment to verify the inverse square law

- Apparatus: small bright lamp/LED (approx. point source with diffuser/pinhole), meter rule, light detector (e.g. calibrated photodiode + ammeter or lux meter), darkened room.
- Method: place detector at distance d from source (measure from the source centre). Record irradiance I (or proportional reading) for several d values, keeping source output constant. Ensure detector face is perpendicular to the beam and at the same height; subtract background light.
- Analysis: show that $I \times d^2$ is (approximately) constant over all data, or plot I against $1/d^2$ to obtain a straight line through the origin.

(b)(i) State what is meant by irradiance

Answer: Power per unit area incident on a surface (W m^{-2}).

(b)(ii) Use all the data to establish the relationship between I and d

Given data:

d (m): 0.200, 0.300, 0.400, 0.500, 0.600

I (W m^{-2}): 142.0, 63.1, 35.5, 22.7, 15.8

Compute $I \times d^2$ for each pair:

- $142.0 \times (0.200)^2 = 142.0 \times 0.0400 = 5.68$
- $63.1 \times (0.300)^2 = 63.1 \times 0.0900 = 5.68$
- $35.5 \times (0.400)^2 = 35.5 \times 0.1600 = 5.68$
- $22.7 \times (0.500)^2 = 22.7 \times 0.2500 = 5.68$
- $15.8 \times (0.600)^2 = 15.8 \times 0.3600 = 5.69$

Since $I \times d^2 \approx \text{constant} (\approx 5.68)$, therefore $I \propto 1/d^2$.

(c) Explain why irradiance decreases with distance

For a point source emitting uniformly, the same power spreads over the surface of a sphere of radius d. Surface area increases as $4\pi d^2$. As d increases, the same power is distributed over a larger area, so power per unit area (irradiance) decreases $\propto 1/d^2$.

Final Answers

(a) Valid apparatus + method to measure I versus d , analyse with $I \times d^2$ constant or I vs $1/d^2$ linear.

(b)(i) Irradiance = power per unit area incident on a surface (W m^{-2}).

(b)(ii) $I \times d^2 \approx 5.68$ (constant) $\rightarrow I \propto 1/d^2$.

(c) Same power spreads over larger spherical area ($\propto d^2$), so I decreases as $1/d^2$.

Revision Tips

- Be explicit: show every $I \times d^2$ product to claim $I \propto 1/d^2$.
- Keep the detector face perpendicular and distance measured from the source centre.
- Inverse square laws also apply to gravity and electric fields for point sources/charges.